

Indian Statistical Institute

End semester examination

Statistics 1: Data Exploration
8 December 2025, 10.30 am – 1.30pm

Bachelor of Statistical Data Science
Total points: 50

INSTRUCTIONS

Write your name and roll number in CAPITAL LETTERS. Write your answers on the question paper itself, within the blank space provided just after question. All pen-paper calculations needs to be done in the space provided, submit only the filled in question paper. Answer all questions.

You are allowed to use a calculator; however, the use of any other electronic device is strictly prohibited.

Name:
Roll number:

1. Consider a sample data set on Monthly Per-Capita Consumption Expenditure (MPCE), derived from Periodic Labor Force Survey (PLFS 2018-19). The following is grouped class interval and approximate frequencies for a sample of 400 rural households.

Monthly MPCE (in rupees)	Frequency
0 – 1,125	40
1,125 – 1,600	60
1,600 – 2,425	110
2,425 – 3,750	90
3,750 – 5,625	50
> 5,625	50

- Compute the median MPCE for the above grouped data (clearly state the formula used and parameters involved).
- Would the mean be higher or lower than the median you compute for this data? why?
- Compute the first and the third quartiles of the MPCE.
- Compute the quartile deviation and interpret the value in the context of variability of the MPCE in rural India.
- Compute a measure of skewness from the above quartiles. Comment on the skewness of MPCE distribution and interpret the result in terms of inequality of the MPCE in rural India. [2+2+2+2+2]

2. (a) Define population Pearson product moment correlation coefficient for two random variable X and Y . Identify the range of its values.
- (b) If X and Y are two random variable with correlation coefficient ρ_{XY} , and we define $V = aX$ and $W = d + cY$, where d is a constant and a and c are positive constant. Suppose V and W has correlation ρ_{VW} . Show that $\rho_{XY} = \rho_{VW}$.
- (c) Let the slope parameter of regression of Y on X is denoted by β_{XY} . Determine the relationship between β_{XY} and ρ_{XY} .
- (d) Suppose X takes values -2, -1, 0, 1, 2 and $Y = X^2$. Compute the correlation between X and Y for this data. Comment on the relationship of the two variables.
- (e) National Family Health Survey (NFHS-5), observed moderate negative correlation between Households with Improved Sanitation and Child Height-for-Age (Stunting), that is, households with improved toilets show lower stunting among children. Do you think, this correlation implies causation, indicating that, having a toilet alone does directly cause better height? Explain your answer and identify any factors that impacts both the above variables. [2+2+2+2+2+2=10]

3. Consider daily air-quality parameter (PM 2.5 in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$)), in RK Puram, Delhi for the first 20 days of November this year as

321, 265, 204, 182, 177, 281, 258, 304, 282, 327, 354, 416, 367, 281, 315, 275, 268, 286, 344, 340.

Data source: www.aqicn.org. Suppose the daily PM 2.5 is represented by a random variable X .

- (a) Define the empirical distribution function (EDF), denoted by $F_n(x)$ of X .
- (b) Compute $F_n(x)$ at $x = 175, 225, 275, 315, 375, 425$.
- (c) Draw a step-function plot of $F_n(x)$ at the sample points: 177, 204, 304, 367, 416 marking clearly where the jumps occur and the size of each jump.
- (d) Use EDF to estimate the probability that PM2.5 lies between $200\mu\text{g}/\text{m}^3$ and $300\mu\text{g}/\text{m}^3$.
- (e) Use EDF to estimate the probability that PM2.5 lies above the sample mean. [2+2+2+2+2=10]

4. Consider the summary data derived from the report of National Family Health Survey (NFHS-5, 2019-21) on years of schooling (X) and women's age of first marriage (Y)

Years of schooling (in years)	Age at first marriage (in years)
0	16.4
3	17
6	17.6
8.5	18.4
10.5	19.3
13	21.3

Let us assume that the above sample data $(Y_i, X_i), i = 1 \dots, 6$, is generated from a simple linear regression model

$$E(Y) = \beta_0 + \beta_1 X, \quad V(Y) = \sigma^2,$$

where β_0 and β_1 are usual regression parameters and σ^2 is error variance.

- Derive in detail, the algebraic form of least square estimates of β_0 and β_1 (denoted by $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively)
- Compute the $\hat{\beta}_0$ and $\hat{\beta}_1$ for the above data.
- Write down the equation of the estimated regression line and; interpret the intercept and slope parameters in the given contexts.
- Using the above data set and assuming $\sigma^2 = 1$, compute numerical values of the variance of estimates (state the form of the variance and full derivation not required).
- Using the above data, describe on graphical method for checking the constant variances using regression residuals.

[2+2+2+2+2=10]

5. (a) Define simple random sampling with replacement (SRSWR) and simple random sample without replacement (SRSWOR) by determining the selection probability of a sample of size n from a population of size N in each case. Give a real life example where SRSWOR would be preferable than SRSWR.
- (b) Suppose a population consists of the five values

$$\{4, 6, 8, 10, 2\}$$

- i. List all possible SRSWOR sample of size 3.
- ii. Find out the selection probability of any sample of size 3 from the population of size 5.
- iii. Using the above sample and population values show that the sample mean is an unbiased estimate of population mean.
- iv. Compute the variance of the sample mean for the above data. [(2)+(2+2+2+2)=10]

